

# Sowing Schedule Planner.

## User Guide & Model Documentation

### 1. What this tool is for?

The Sowing Schedule Planner helps growers prepare a sowing sequence ahead of a sowing season, deciding what to sow, where (which frost-risk zone), and when, to make the best use of a farm's paddocks, machinery capacity, and crop targets.

This tool is intended for pre-season planning, not tactical, in-season decision-making. It is designed to be used ahead of time, so a grower walks into the season with a clear, considered plan rather than making sowing-order decisions on the fly.

Beyond producing a single recommended plan, the tool is also useful for exploring “what if” scenarios, such as:

- What if a piece of sowing equipment broke down, reducing daily capacity?
- What if commodity prices for a crop were unusually high or low?
- What if I used sowing equipment with a different daily capacity?
- What if I was willing to sow any crop in a high-frost-risk zone, rather than excluding one?
- What combination of crops and sowing sequence makes the best use of the yield potential across my paddocks?

### 2. How the optimiser works

#### 2.1 Two conceptual approaches

There are two broad ways a sowing schedule could be built. It is useful to understand both, because it explains why this tool takes the approach it does.

##### Priority-based (greedy) scheduling

A simple, rule-based approach: in each sowing week, rank the available crops and zones by that week's yield, and allocate all available capacity to the top-ranked option first, spilling any leftover into the next-ranked option.

This method is transparent and easy to explain (“we always plant whatever is best this week”), but it makes its decision one week at a time, in isolation, with no visibility into future weeks. It cannot trade off across weeks, for example, sacrificing a good-but-not-best week now in order to preserve capacity for an even better week later, because it never looks beyond the current week's comparison.

##### Whole-season optimisation (the approach used by this tool)

Rather than deciding week by week, the tool considers every sowing week and every crop simultaneously, as one combined problem. It searches for the single allocation of hectares, across every crop, zone, and week, that produces the best total outcome for the whole season at once, subject to the farm's real constraints.

Because it evaluates the whole season together, it can produce plans that a week-by-week rule has no mechanism to find. For example, skipping a mediocre week for one crop entirely, in order to concentrate that crop into its genuinely strongest windows elsewhere in the program.

## 2.2 Formal statement of the model

The scheduling problem is formulated as a Mixed-Integer Linear Program (MILP).

Let:

- $c$  index the set of crops,
- $z$  index the set of frost-risk zones (Green, Amber, Red),
- $w$  index the set of sowing weeks in the program,
- $ha_{c,z,w}$  be the decision variable: hectares of crop  $c$  sown in zone  $z$  in week  $w$  (continuous,  $\geq 0$ ).

Decision variables also include binary indicators tracking whether a crop is actively being sown in a given week, and whether a crop is starting in a given week (used to enforce the sequencing rule described in Section 2.4).

### Objective 1. Maximise expected yield

$$\text{maximise } \sum_c \sum_z \sum_w y_{c,z,w} \times ha_{c,z,w}$$

where  $y_{c,z,w}$  (yield, t/ha) is written as  $y$  with subscript  $c,z,w$  — the expected yield for crop  $c$  sown in zone  $z$  in week  $w$ , under the season type currently being evaluated (see Section 2.3).

### Objective 2. Maximise expected gross margin (alternative)

The tool also supports optimising for expected gross margin instead of yield:

$$\text{maximise } \sum_c \sum_z \sum_w (y_{c,z,w} \times \text{price}_c - vc_{c,\text{decile}}) \times ha_{c,z,w}$$

where  $\text{price}_c$  (\$/t) is the grain price for crop  $c$ , and  $vc_{c,\text{decile}}$  (\$/ha) is the variable cost of production for crop  $c$  under the current season type.

### Subject to

- Crop targets: total hectares sown of each crop must equal the user-specified target area for that crop.
- Weekly capacity: total hectares sown across all crops and zones in any single week cannot exceed that week's sowing capacity.
- Zone capacity: total hectares sown in each frost-risk zone across the whole program cannot exceed that zone's available area.
- Zone exclusion: optionally, one nominated crop may be excluded entirely from the Red (highest frost-risk) zone.
- Non-negativity: all hectares sown must be  $\geq 0$ .
- Sequencing constraints: see Section 2.4.

### 2.3 Season types (decile groups)

The optimiser can be run separately for three season types, representing a poor, average, and good season respectively: D1–3, D4–6, and D7–9. These groupings are based on historical yield deciles (not rainfall deciles). That is, years are grouped by how the resulting crop yield actually ranked historically, not by how much rain fell. The yields used throughout are real, observed crop yields, not raw crop-model (e.g. APSIM) output.

Running the optimiser once per season type shows how the recommended plan, and its expected outcome, changes depending on the kind of season that eventuates.

### 2.4 The sowing sequencing rule (“finish before you switch”)

The model reflects how growers actually operate: working through one crop at a time, rather than jumping back and forth between crops.

In plain terms: one crop has exclusive claim on all available sowing capacity until its entire target area has been sown. A second crop can only begin once the first is completely finished, with one exception. If finishing the first crop does not use up all of that week's capacity, the next crop may use the leftover capacity in that same week (a “handover week”). The outgoing crop must be fully complete by the end of that week.

Formally, for any two crops  $c_1 \neq c_2$ , if crop  $c_2$  starts sowing in week  $w$ , then crop  $c_1$  (if it has already begun sowing at any point up to week  $w$ ) must have already reached its full target area by the end of week  $w$ :

$$\sum_z \sum_{w' \leq w} ha_{c_1, z, w'} \geq target_{c_1} \times (start_{c_2, w} + started_{c_1, w} - 1)$$

where  $started_{c_1, w}$  is 1 if crop  $c_1$  has started sowing at or before week  $w$ , and 0 otherwise. This constraint only binds when both crop  $c_2$  is starting in week  $w$  and crop  $c_1$  has already begun — it does not restrict a crop that has not yet started at all.

## 3. How to use the tool

### 3.1 Setup tab

Enter your farm and scenario details:

- Site name, simulation name, cropping area, and daily sowing capacity.
- Frost zone split (% of the farm in Green, Amber, and Red zones).
- Crop targets, hectares of Wheat, Barley, Canola, and up to two legume crops.
- An optional single crop to exclude from the Red (highest-risk) zone.
- Program start date.
- Grain price scenario, and whether to optimise for Yield or Gross Margin.
- The yield input file to use (a bundled preset, or your own uploaded file).

The page shows a live feasibility check, ie confirming there is enough sowing capacity for your crop targets given the program start date, before you click Run.

### 3.2 Report tab

After running a simulation, this tab shows the recommended sowing plan for each of the three season types (D1–3, D4–6, D7–9) side by side — a timeline of what to sow, where, and when — along with expected yield and expected gross margin for each. A button lets you download the full report and underlying data as a single .zip file.

### 3.3 Compare tab

Upload two or more previously downloaded simulation bundles (.zip files) to compare their settings and outcomes side by side. This is useful for reviewing multiple scenarios you have already run, without needing to re-run them.